

UAI402: Data Engineering

L	T	P	Cr
3	0	2	4.0

Course Objectives: This course aims to explore the fundamentals and applications of data engineering, with a consideration of sustainability. Students will acquire the ability to minimize their environmental impact by designing, constructing, and maintaining data systems. The subjects covered include data lifecycle management, green computation, data pipelines, and storage optimization.

Foundations of Data Engineering and Data Modeling: Introduction to Data Engineering: Roles, workflows, and modern data stack, Data types, file formats (CSV, JSON, Avro, Parquet), Batch vs. streaming data, Data modeling: ER diagrams, normalization, star/snowflake schema, Introduction to data warehousing and OLAP, Use case: data modelling for 1. student performance, 2. school access in rural areas ⁴

ETL/ELT Pipelines and Workflow Orchestration: Designing ETL/ELT pipelines, Data ingestion from APIs, sensors, and flat files, Data transformation and cleansing techniques, Workflow orchestration: Apache Airflow, Prefect, Error handling, retries, and monitoring, Use case: Ingesting and processing sensor data from water quality monitoring stations for real-time alerts and long-term reporting ⁶

Distributed Data Processing and Big Data Frameworks: Introduction to distributed systems and Apache Hadoop, Apache Spark: RDDs, DataFrames, Spark SQL, Partitioning, shuffling, and performance optimization, Batch vs. stream processing fundamentals, Use case: Analyzing large-scale satellite and weather data to track deforestation and CO₂ emissions over time using Spark¹³

Real-Time Data Streaming and Messaging Systems: Introduction to message brokers and stream processing, Apache Kafka: architecture and core concepts, Stream processing with Spark Streaming / Apache Flink, Event time, windows, watermarks, Use case: Real-time energy consumption tracking from IoT meters to optimize grid load and promote renewable energy usage ⁷

Data Governance, Cloud Deployment, and DevOps: Data quality and validation: expectations and checks, Metadata, lineage, and cataloguing, Security, encryption, and compliance (GDPR, HIPAA), Cloud platforms: AWS/GCP/Azure basics, Docker, Terraform, CI/CD pipelines for data workflows, Use case: Deploying a data platform in the cloud to monitor and visualize real-time traffic and pollution levels in urban areas¹¹

Laboratory work:

- Creating and monitoring ETL pipelines
- Ingesting real-time sensor data using Kafka
- Running distributed processing jobs in Apache Spark
- Using Airflow for orchestration and scheduling
- Validating data quality and deploying pipelines on cloud infrastructure

Course Outcomes (COs)/Course Learning Outcomes (CLOs):

After completion of this course, the students will be able to:

1. Understand and design scalable data models and workflows for structured and unstructured data.
2. Build and manage ETL/ELT pipelines using modern tools and automation frameworks.
3. Apply distributed and real-time processing techniques to solve large-scale data problems.
4. Develop data engineering solutions for real-world challenges, including sustainability, governance, and cloud deployment.

Text Books:

1. Andreas Kretz, The Data Engineering Cookbook, 2022
2. Ananthakrishnan G., Data Engineering with Python, Packt Publishing, 2020

Reference Books:

1. Jesse Anderson, Data Engineering Teams, Manning, 2020
2. Jeffrey Aven, Data Lake Architecture, O'Reilly Media, 2018
3. Bill Inmon, Building the Data Warehouse, Wiley, 4th ed.
4. Valliappa Lakshmanan et al., Data Engineering on Google Cloud Platform, O'Reilly, 2022